

3.4 Calculus	understand the concept of a variable rate of change differentiate integer powers of x determine gradients, rates of change, turning points (maxima and minima) by differentiation and relate these to graphs distinguish between maxima and minima by considering the general shape of the graph apply calculus to linear kinematics and to other simple practical problems	$y = x + \frac{9}{x}$ Find the coordinates of the maximum and minimum points The displacement, s metres, of a particle from a fixed point O after t seconds is given by: $s = 24t^2 - t^3,$ $0 \leq t \leq 20.$ Find expressions for the velocity and the acceleration.
---------------------	---	---

A02 Shape, space and measures		
4 Geometry		
	Students should be taught to:	Notes
4.1 Lines and triangles	See Foundation Tier.	
4.2 Polygons	See Foundation Tier.	
4.3 Symmetry	See Foundation Tier.	
4.4 Measures	See Foundation Tier.	
4.5 Construction	See Foundation Tier.	
4.6 Circle properties	understand and use the internal and external intersecting chord properties recognise the term <i>cyclic quadrilateral</i> understand and use angle properties of the circle including: - angle subtended by an arc at the centre of a circle is twice the angle subtended at any point on the remaining part of the circumference - angle subtended at the circumference by a diameter is a right angle - angles in the same segment are equal - the sum of the opposite angles of a cyclic quadrilateral is 180° - the alternate segment theorem	Formal proof of these theorems is not required
4.7 Geometrical reasoning	provide reasons, using standard geometrical statements, to support numerical values for angles obtained in any geometrical context involving lines, polygons and circles	
4.8 Trigonometry and Pythagoras' Theorem	understand and use sine, cosine and tangent of obtuse angles understand and use angles of elevation and depression understand and use the sine and cosine rules for any triangle use Pythagoras' Theorem in 3 dimensions understand and use the formula $\frac{1}{2}bc\sin C$ for the area of a triangle	

	apply trigonometrical methods to solve problems in 3 dimensions, including finding the angle between a line and a plane	The angle between two planes will not be required
--	--	--